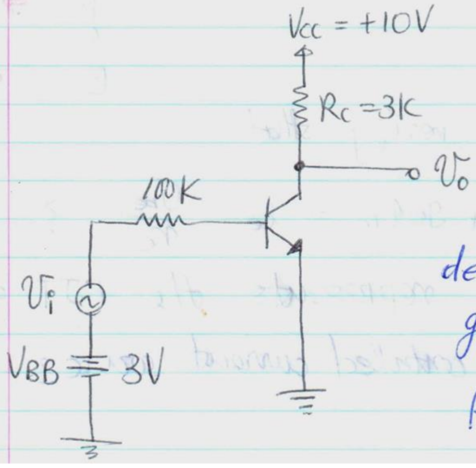
Problems on slides

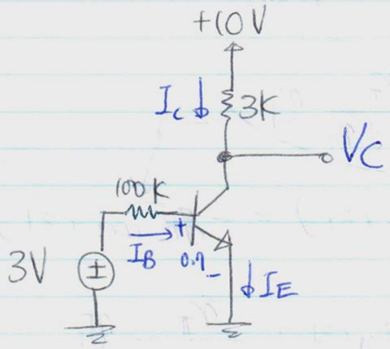
⑩ Example.



determine the voltage gain, v_o/v_i

$\beta = 100$.

i) DC-bias analysis



$I_B = \frac{3-0.7}{100K}$

$= 0.023\ mA$

$I_C = \beta \cdot I_B = 2.3\ mA$

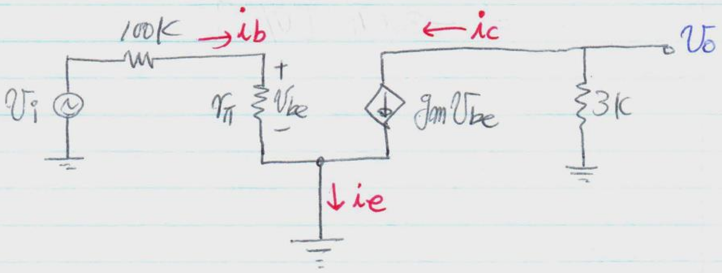
$I_E = I_B + I_C = 2.323\ mA$

$V_C = 10 - (3K)\,I_C$

$= +\ 3.1\ V$

$\Rightarrow V_C > 0.3V \Rightarrow$ "active mode"

ii) small-signal analysis



$\cdot\ g_m = \frac{I_C}{V_T} = \frac{2.3m}{0.025} = 92\ mA/V$

要注意 Vt

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{92m} = 1.09\ k\Omega$$

$$v_{be} = \frac{r_\pi}{100k + r_\pi} v_i$$

$$v_o = -i_c(3k) = -g_m v_{be}(3k)$$

$$\Rightarrow v_o = -g_m(3k)\frac{r_\pi}{100k + r_\pi} v_i$$

$$\Rightarrow A_v = \frac{v_o}{v_i} = -g_m(3k)\frac{r_\pi}{100k + r_\pi}$$

$$= -(92m)(3k)\frac{1.09k}{100k + 1.09k}$$

$$= -3.04\ [V/V]$$

要注意这里的 vi 不是普通的 vbe，需要进行转换

Problem on textbook

Chapter 4

D *4.78 It is required to use a peak rectifier to design a dc power supply that provides an average dc output voltage of 12 V on which a maximum of ±1-V ripple is allowed. The rectifier feeds a load of 200 Ω. The rectifier is fed from the line voltage (120 V rms, 60 Hz) through a transformer. The diodes available have 0.7-V drop when conducting. If the designer opts for the half-wave circuit:

(a) Specify the rms voltage that must appear across the transformer secondary.

(b) Find the required value of the filter capacitor.

(c) Find the maximum reverse voltage that will appear across the diode, and specify the PIV rating of the diode.

(d) Calculate the average current through the diode during conduction.

(e) Calculate the peak diode current.

∨ **Show Answer**

D *4.79 Repeat Problem 4.78 for the case in which the designer opts for a full-wave circuit utilizing a center-tapped transformer.

D *4.80 Repeat Problem 4.78 for the case in which the designer opts for a full-wave bridge-rectifier circuit.

Problem 4.78

A half-wave peak rectifier must provide an average dc output voltage of 12 V with maximum ripple ±1 V. Therefore

$$V_r = 2\,\text{V} \quad \text{peak-to-peak}, \qquad V_{O,avg} = 12\,\text{V}, \qquad R = 200\,\Omega, \qquad I_{dc} = \frac{12}{200} = 60\,\text{mA}. \tag{37}$$

Since the average is midway between the top and bottom of the ripple,

$$V_M = V_{O,max} = V_{O,avg} + \frac{V_r}{2} = 12 + 1 = 13\,\text{V}. \tag{38}$$

For the half-wave rectifier, only one diode conducts, so

$$V_{S,p} = V_M + V_D = 13 + 0.7 = 13.7\,\text{V}. \tag{39}$$

(a) Transformer secondary rms voltage

$$V_{sec,rms} = \frac{V_{S,p}}{\sqrt{2}} = \frac{13.7}{\sqrt{2}} = 9.69\,\text{V}_{\text{rms}}. \tag{40}$$

(b) Filter capacitor

For a half-wave peak rectifier, the capacitor discharges for approximately one full line period:

$$V_r \simeq \frac{I_{dc}}{fC} \quad \Rightarrow \quad C \simeq \frac{I_{dc}}{fV_r} = \frac{0.060}{60(2)} = 5.00 \times 10^{-4}\,\text{F}. \tag{41}$$

Thus

$$\boxed{C \simeq 500\,\mu\text{F}}. \tag{42}$$

(c) Maximum reverse voltage and PIV rating

At the negative peak of the source, the diode sees approximately the charged capacitor voltage plus the negative source magnitude:

$$V_{R,max} \simeq V_{S,p} + V_M = 13.7 + 13 = 26.7\,\text{V}. \tag{43}$$

Thus the minimum PIV rating is approximately 26.7 V. A practical choice would be at least 30 V, and a standard 50 V diode rating would be safer.

(d) Average diode current during conduction

$$\frac{\Delta t}{T} \simeq \frac{1}{2\pi}\sqrt{\frac{2V_r}{V_{S,p}}} = \frac{1}{2\pi}\sqrt{\frac{4}{13.7}} = 0.0860, \tag{44}$$

$$I_{D,avg|cond} \simeq I_{dc}\frac{T}{\Delta t} = \frac{0.060}{0.0860} = 0.698\,\text{A}. \tag{45}$$

注意 PIV 的计算需要精确，sliides 中提到的只是近似值

(e) Peak diode current

Approximating the charging pulse as triangular,

$$I_{D,peak} \simeq 2I_{D,avg|cond} = 1.40\,\text{A}. \tag{46}$$

Quantity	Result
$V_{sec,rms}$	$9.69\,\text{V}_{\text{rms}}$
C	$500\,\mu\text{F}$
$V_{R,max}$ / minimum PIV	$26.7\,\text{V}$; choose $\geq 30\,\text{V}$, preferably $50\,\text{V}$
$I_{D,avg\|cond}$	$0.698\,\text{A}$
$I_{D,peak}$	$1.40\,\text{A}$

Problem 4.79

This repeats Problem 4.78 for a center-tapped full-wave rectifier.

(a) Transformer secondary rms voltage

Only one diode conducts in each charging path, so the required half-secondary peak is the same as in Problem 4.78:

$$V_{S,p} = V_M + V_D = 13.7\,\text{V}. \tag{47}$$

Therefore

$$V_{half,rms} = \frac{13.7}{\sqrt{2}} = 9.69\,\text{V}_{\text{rms}}. \tag{48}$$

Because the transformer is center tapped, the total secondary voltage is

$$V_{sec,total,rms} = 2(9.69) = 19.4\,\text{V}_{\text{rms}} \quad \text{end-to-end.} \tag{49}$$

(b) Filter capacitor

The capacitor is recharged every half-cycle, so the discharge interval is approximately $T/2$:

$$C \simeq \frac{I_{dc}}{(2f)V_r} = \frac{0.060}{120(2)} = 2.50 \times 10^{-4}\,\text{F}. \tag{50}$$

Thus

$$\boxed{C \simeq 250\,\mu\text{F}}. \tag{51}$$

(c) Maximum reverse voltage and PIV rating

For a center-tapped full-wave rectifier, the nonconducting diode sees roughly the charged capacitor voltage plus the opposite half-secondary peak:

$$V_{R,max} \simeq V_{S,p} + V_M = 13.7 + 13 = 26.7\,\text{V}. \tag{52}$$

Thus the minimum PIV rating is approximately $26.7\,\text{V}$. Use at least $30\,\text{V}$; $50\,\text{V}$ is a safer standard rating.

(d) Average diode current during conduction

The conduction fraction of each diode is still set by the sinusoidal peak curvature:

$$\frac{\Delta t}{T} = 0.0860. \tag{53}$$

However, each diode supplies only one half-cycle of load charge:

$$I_{D,avg|cond} \simeq I_{dc}\frac{T/2}{\Delta t} = \frac{0.060}{2(0.0860)} = 0.349\,\text{A}. \tag{54}$$

(e) Peak diode current

$$I_{D,peak} \simeq 2(0.349) = 0.698\,\text{A}. \tag{55}$$

Quantity	Result
Half-secondary rms voltage	$9.69\,\text{V}_{\text{rms}}$
Total secondary rms voltage	$19.4\,\text{V}_{\text{rms}}$ end-to-end
C	$250\,\mu\text{F}$
$V_{R,max}$ / minimum PIV	$26.7\,\text{V}$; choose $\geq 30\,\text{V}$, preferably $50\,\text{V}$
$I_{D,avg\|cond}$	$0.349\,\text{A}$
$I_{D,peak}$	$0.698\,\text{A}$

Full-wave bridge 的需要两倍的 Von，PIV 也有一点区别，需要根据题目分析

Problem 4.80

This repeats Problem 4.78 for a full-wave bridge rectifier.

(a) Transformer secondary rms voltage

A bridge rectifier has two conducting diodes in series, so

$$V_{S,p} = V_M + 2V_D = 13 + 1.4 = 14.4\,\mathrm{V}. \tag{56}$$

Thus

$$V_{sec,rms} = \frac{14.4}{\sqrt{2}} = 10.18\,\mathrm{V_{rms}}. \tag{57}$$

(b) Filter capacitor

The bridge rectifier recharges the capacitor every half-cycle:

$$C \simeq \frac{I_{dc}}{(2f)V_r} = \frac{0.060}{120(2)} = 250\,\mu\mathrm{F}. \tag{58}$$

(c) Maximum reverse voltage and PIV rating

For the bridge rectifier, the required PIV is approximately the secondary peak voltage:

$$V_{R,max} \simeq V_{S,p} = 14.4\,\mathrm{V}. \tag{59}$$

With the constant diode-drop model, one may also view the reverse voltage as about $V_M + V_D = 13.7\,\mathrm{V}$. The standard design answer is therefore approximately 14.4 V; choose a diode with PIV at least 15 V, practically 20–50 V.

(d) Average diode current during conduction

The conduction fraction is

$$\frac{\Delta t}{T} \simeq \frac{1}{2\pi}\sqrt{\frac{2V_r}{V_{S,p}}} = \frac{1}{2\pi}\sqrt{\frac{4}{14.4}} = 0.0839. \tag{60}$$

Each diode in the bridge conducts once per line cycle and carries one half-cycle of load charge:

$$I_{D,avg|cond} \simeq I_{dc}\frac{T/2}{\Delta t} = \frac{0.060}{2(0.0839)} = 0.358\,\mathrm{A}. \tag{61}$$

(e) Peak diode current

$$I_{D,peak} \simeq 2(0.358) = 0.715\,\mathrm{A}. \tag{62}$$

Quantity	Result
$V_{sec,rms}$	$10.18\,\mathrm{V_{rms}}$
C	$250\,\mu\mathrm{F}$
$V_{R,max}$ / minimum PIV	≈ 14.4 V; choose ≥ 15 V, practically 20–50 V
$I_{D,avg\vert cond}$	0.358 A
$I_{D,peak}$	0.715 A

Chapter 6

6.2 Two transistors, fabricated with the same technology but having different junction areas, when operated at a base-emitter voltage of 0.75 V, have collector currents of 0.4 mA and 2 mA. Find I_S for each device. What are the relative junction areas?

∧ **Hide Answer**

3.7×10^{-17} A; 1.87×10^{-16} A; 5:1

Problem 6.2

The active-mode collector current is

$$I_C = I_S e^{V_{BE}/V_T}.$$

Given $V_{BE} = 30V_T$, the scale current is

$$I_S = I_C e^{-30}.$$

For $I_C = 0.4\,\mathrm{mA}$,

$$I_{S1} = 0.4\,\mathrm{mA}e^{-30} = 3.74 \times 10^{-17}\,\mathrm{A}.$$

For $I_C = 2\,\mathrm{mA}$,

$$I_{S2} = 2\,\mathrm{mA}e^{-30} = 1.87 \times 10^{-16}\,\mathrm{A}.$$

Since I_S is proportional to emitter-base junction area,

$$\frac{A_2}{A_1} = \frac{I_{S2}}{I_{S1}} = 5.$$

$$\boxed{I_{S1} = 3.74 \times 10^{-17}\,\mathrm{A}, \quad I_{S2} = 1.87 \times 10^{-16}\,\mathrm{A}, \quad A_2/A_1 = 5}$$

知识点： I_S 不是普通“电源电流”，它是器件尺寸和工艺决定的 scale current；面积越大，同样 V_{BE} 下电流越大。

I_s 和 A 呈正比例

***6.11** For a transistor with α close to unity, show that if α changes by a small per-unit amount $(\Delta\alpha/\alpha)$, the corresponding per-unit change in β is given approximately by

$$\frac{\Delta\beta}{\beta} \simeq \beta\left(\frac{\Delta\alpha}{\alpha}\right)$$

Now, for a transistor whose nominal β is 100, find the percentage change in its α value corresponding to a drop in its β of 10%.

最简正确写法

$$\alpha = \frac{\beta}{\beta+1}$$

$$\frac{d\alpha}{d\beta} = \frac{1}{(\beta+1)^2}$$

$$\frac{d\alpha}{\alpha} = \frac{1}{\alpha(\beta+1)^2}d\beta$$

因为：

$$\alpha = \frac{\beta}{\beta+1}$$

所以：

$$\alpha(\beta+1)^2 = \beta(\beta+1)$$

因此：

$$\frac{d\alpha}{\alpha} = \frac{d\beta}{\beta(\beta+1)}$$

$$\boxed{\frac{d\alpha}{\alpha} = \frac{1}{\beta+1}\frac{d\beta}{\beta}}$$

当 $\beta \gg 1$：

$$\boxed{\frac{d\alpha}{\alpha} \approx \frac{1}{\beta}\frac{d\beta}{\beta}}$$

注意近似

D **6.32** Design the circuit in Fig. P6.32 to establish a current of 0.5 mA in the emitter and a voltage of −0.5 V at the collector. The transistor v_{EB} = 0.7 V at I_E = 1 mA, and β = 100. To what value can R_C be increased while the collector current remains unchanged?

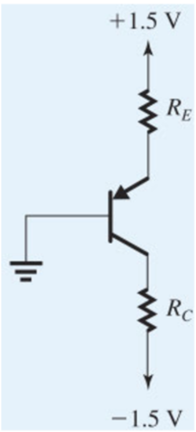


Figure P6.32

Problem 6.32

For the pnp circuit, $I_E = 0.5\,\text{mA}$ and $\beta = 100$. At $I_E = 0.5\,\text{mA}$,

$$V_{EB} = 0.700 + V_T \ln \frac{0.5mA}{1mA} = 0.683\,\text{V}.$$

Thus

$$V_E = +0.683\,\text{V}.$$

The emitter resistor is

$$R_E = \frac{1.5 - 0.683}{0.5mA} = 1.63\,\text{k}\Omega.$$

The collector current is

$$I_C = \frac{100}{101}(0.5mA) = 0.495\,\text{mA}.$$

To set $V_C = -0.5\,\text{V}$,

$$R_C = \frac{-0.5 - (-1.5)}{0.495mA} = 2.02\,\text{k}\Omega.$$

The largest R_C before saturation occurs at approximately $V_C \approx V_B + 0.4 = 0.4\,\text{V}$, so

$$R_{C,\max} = \frac{0.4 - (-1.5)}{0.495mA} = 3.84\,\text{k}\Omega.$$

$$\boxed{R_E = 1.63\,k\Omega,\ R_C = 2.02\,k\Omega,\ R_{C,\max} \approx 3.84\,k\Omega}$$

知识点： pnp active 要求 collector 电位低于 base 电位；R_C 太大时 collector 被拉高，晶体管进入饱和。

注意这个 0.7 只是近似值，只能用于检查是否位于饱和区，不能直接带入，只能在题目说是近似模型的时候使用

6.40 Consider a transistor for which the base–emitter voltage drop is 0.7 V at 10 mA. What current flows for v_{BE} = 0.5 V? Evaluate the ratio of the slopes of the i_C–v_{BE} curve at v_{BE} = 700 mV and at v_{BE} = 500 mV. The large ratio confirms the point that the BJT has an "apparent threshold" at $v_{BE} \simeq 0.5$ V.

Problem 6.40

For $I_C = 10\text{mA}$ at $V_{BE} = 0.7\text{V}$, at $V_{BE} = 0.5\text{V}$,

$$I_C = 10mA\, e^{(0.5-0.7)/0.025} = 3.35\,\mu\text{A}.$$

The slope of the transfer characteristic is

$$\frac{dI_C}{dV_{BE}} = \frac{I_C}{V_T}.$$

Thus the slope ratio between the two operating points is the current ratio:

$$\frac{(dI_C/dV_{BE})_{0.7}}{(dI_C/dV_{BE})_{0.5}} = \frac{10mA}{3.35\mu A} = 2.98 \times 10^3.$$

$$\boxed{I_C(0.5V) = 3.35\,\mu A, \quad \text{slope ratio} \approx 2.98 \times 10^3}$$

知识点： $g_m = dI_C/dV_{BE} = I_C/V_T$，所以电流越大，小信号跨导越大。

两个点的斜率分别是：

$$g_{m1} = \frac{I_{C1}}{V_T}$$

$$g_{m2} = \frac{I_{C2}}{V_T}$$

所以斜率比：

$$\frac{g_{m1}}{g_{m2}} = \frac{I_{C1}/V_T}{I_{C2}/V_T}$$

V_T抵消：

$$\boxed{\frac{g_{m1}}{g_{m2}} = \frac{I_{C1}}{I_{C2}}}$$

6.48 For the circuit in Fig. P6.48 let $V_{CC} = 5$ V, $R_C = 10$ kΩ, and $R_B = 100$ kΩ. The BJT has $\beta = 50$. Find the value of V_{BB} that results in the transistor operating

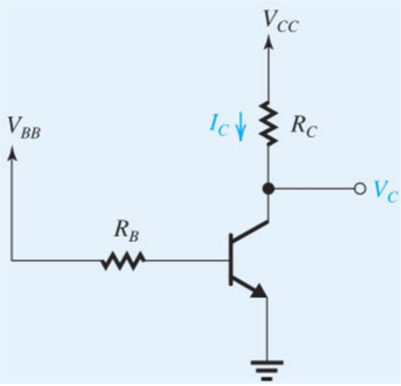


Figure P6.48

(a) in the active mode with $V_C = 2$ V;

(b) at the edge of saturation;

(c) deep in saturation with $\beta_{forced} = 10$.

Assume $V_{BE} \simeq 0.7$ V.

Problem 6.48

For the npn circuit, use $V_{BE} \approx 0.7$V and $\beta = 50$.

(a) To obtain $V_C = 2$V,

$$I_C = \frac{5-2}{10k} = 0.300\,\text{mA}, \quad I_B = \frac{I_C}{50} = 6.0\,\mu\text{A}.$$

Thus

$$V_{BB} = 0.7 + I_B(100k) = 1.30\,\text{V}.$$

(b) At the edge of saturation, take $V_{CE} \approx 0.3$V:

$$I_C = \frac{5-0.3}{10k} = 0.470\,\text{mA}, \quad I_B = \frac{0.470mA}{50} = 9.4\,\mu\text{A}.$$

$$V_{BB} = 0.7 + (9.4\mu A)(100k) = 1.64\,\text{V}.$$

(c) For deep saturation with $\beta_f = 10$ and $V_{CE} \approx 0.2$V,

$$I_C = \frac{5-0.2}{10k} = 0.480\,\text{mA}, \quad I_B = \frac{I_C}{10} = 48\,\mu\text{A}.$$

$$V_{BB} = 0.7 + (48\mu A)(100k) = 5.50\,\text{V}.$$

$$\boxed{V_{BB} = 1.30\,V,\ 1.64\,V,\ 5.50\,V}$$

知识点： 判定饱和可以看 forced $\beta = I_C/I_B$ 是否小于器件 β。

题目中说了近似为 0.7 就可以用近似模型

深饱和区域要用 β forced

情况	**常用近似**
BE junction 正向导通	$V_{BE} \approx 0.7V$
edge of saturation 刚饱和	$V_{CE} \approx 0.3V$
deep saturation 深饱和	$V_{CE} \approx 0.2V$

SIM **D *6.49** Consider the circuit of Fig. P6.48 for the case $V_{BB} = V_{CC}$. If the BJT is saturated, use the equivalent circuit of Fig. 6.21 to derive an expression for β_{forced} in terms of V_{CC} and (R_B/R_C). Also derive an expression for the total power dissipated in the circuit. For $V_{CC} = 5$V, design the circuit so that it operates at a forced β as close to 10 as possible while limiting the power dissipation to no larger than 20 mW. Use 1% resistors (see Appendix J).

Problem 6.49

With $V_{BB} = V_{CC}$ and saturation,

$$\beta_f = \frac{I_C}{I_B} = \frac{(V_{CC} - V_{CE\text{sat}})/R_C}{(V_{CC} - V_{BE})/R_B} = \frac{R_B}{R_C}\frac{V_{CC} - V_{CE\text{sat}}}{V_{CC} - V_{BE}}.$$

The power drawn is

$$P = V_{CC}\left(\frac{V_{CC} - V_{CE\text{sat}}}{R_C} + \frac{V_{CC} - V_{BE}}{R_B}\right).$$

For $V_{CC} = 5$V, $V_{BE} = 0.7$V, $V_{CE\text{sat}} = 0.2$V, and target $\beta_f \approx 10$,

$$\frac{R_B}{R_C} = 10\frac{5 - 0.7}{5 - 0.2} = 8.96.$$

Choosing $R_C = 1.33\,k\Omega$ and $R_B = 11.8\,k\Omega$ gives

$$\beta_f = 9.90,$$

$$P = 5\left(\frac{4.8}{1.33k} + \frac{4.3}{11.8k}\right) = 19.9\,\text{mW}.$$

$$\boxed{R_C \approx 1.33\,k\Omega, \quad R_B \approx 11.8\,k\Omega, \quad P \approx 19.9\,mW}$$

知识点：开关设计常用 forced β，不是器件标称 β。强迫较小 β_f 可以确保饱和，但会浪费 base drive。

注意功率计算不能只算电阻还要包含 BJT，所以需要使用 P＝UI 公式来计算

6.50 The *pnp* transistor in the circuit in Fig. P6.50 has $\beta = 50$. Show that the BJT is operating in the saturation mode and find β_{forced} and V_C. To what value should R_B be increased in order for the transistor to operate at the edge of saturation?

Problem 6.50

For the pnp circuit, $V_E = +3$V and $V_B \approx 2.3$V. Thus

$$I_B = \frac{2.3}{10k} = 0.230\,\text{mA}.$$

If the transistor were active with $\beta = 50$,

$$I_C = 50 I_B = 11.5\,\text{mA},$$

which is impossible through a 1 $k\Omega$ collector resistor connected to ground. Therefore the transistor is saturated. With $V_{EC\text{sat}} \approx 0.2$V,

$$V_C \approx V_E - 0.2 = 2.8\,\text{V}, \quad I_C = \frac{2.8}{1k} = 2.8\,\text{mA}.$$

The forced gain is

$$\beta_f = \frac{2.8mA}{0.230mA} = 12.2.$$

At the edge of saturation, use $V_{EC} \approx 0.3$V, giving

$$I_C \approx \frac{2.7}{1k} = 2.7\,\text{mA}, \quad I_B = \frac{I_C}{50} = 54\,\mu\text{A}.$$

Then

$$R_B = \frac{2.3}{54\mu A} = 42.6\,\text{k}\Omega.$$

$$\boxed{V_C \approx 2.8\,V,\ I_C \approx 2.8\,mA,\ \beta_f \approx 12.2,\ R_{B,edge} \approx 42.6\,k\Omega}$$

知识点：如果 active 假设算出的 collector current 超过外部电阻允许值，active 假设立即失败，必须转入饱和分析。

状态	常用电压	是否用正常 β	典型判断
Active 主动区	$V_{BE} \approx 0.7V$	用 $I_C = \beta I_B$	$V_C > V_B > V_E$ for NPN
Edge of saturation 饱和边缘	$V_{CE} \approx 0.3V$	仍用正常 β 作边界	$I_C \approx \beta I_B$
Deep saturation 深饱和	$V_{CEsat} \approx 0.2V$	不用正常 β	$\beta_f = I_C/I_B < \beta$

***6.53** Consider the operation of the circuit shown in Fig. P6.53 for V_B at –1 V, 0 V, and +1 V. Assume that β is very high. What values of V_E and V_C result? At what value of V_B does the emitter current reduce to one-tenth of its value for $V_B = 0$ V? For what value of V_B is the transistor just at the edge of conduction? ($v_{BE} = 0.5$ V) What values of V_E and V_C correspond? For what value of V_B does the transistor reach the edge of saturation? What values of V_C and V_E correspond? Find the value of V_B for which the transistor operates in saturation with a forced β of 1.5.

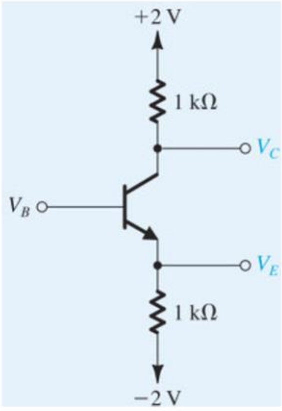


Figure P6.53

Problem 6.53

For the npn circuit with $R_C = R_E = 1\,k\Omega$ and supplies ± 2V, in active mode:

$$V_E = V_B - 0.7, \quad I_E \approx \frac{V_B + 1.3}{1k}, \quad V_C = 2 - I_C(1k) \approx 0.7 - V_B.$$

V_B	V_E	V_C	Region
-1.0	-1.7	$+1.7$	active
0	-0.7	$+0.7$	active
$+1.0$	$+0.3$	$+0.5$	saturation, using $V_{CEsat} \approx 0.2V$

At $V_B = 0$, $I_E = 1.3$mA. One-tenth of this current is 0.13mA:

$$V_E = -2 + 0.13 = -1.87\,\text{V}, \quad V_B = V_E + 0.7 = -1.17\,\text{V}.$$

At cutoff edge, take $V_{BE} \approx 0.5$V and $V_E \approx -2$V:

$$V_B \approx -1.5V.$$

At the edge of saturation, set $V_C = V_B$ in the active formulas:

$$0.7 - V_B = V_B \Rightarrow V_B = 0.35\,\text{V}.$$

For forced $\beta = 1.5$ and $V_{CEsat} \approx 0.2$V,

$$V_E = V_B - 0.7, \quad V_C = V_B - 0.5.$$

Then

$$I_C = \frac{2 - (V_B - 0.5)}{1k} = (2.5 - V_B)mA,$$

$$I_E = \frac{V_B - 0.7 + 2}{1k} = (V_B + 1.3)mA, \quad I_B = I_E - I_C = (2V_B - 1.2)mA.$$

Set $I_C/I_B = 1.5$:

$$\frac{2.5 - V_B}{2V_B - 1.2} = 1.5 \Rightarrow V_B = 1.075\,\text{V}.$$

知识点： 同一电路可随着 V_B 从 cutoff 到 active 再到 saturation。边界点用 junction 条件或 $V_C = V_B$ 这种物理条件找。

$$\boxed{i_C = I_S e^{v_{BE}/V_T}}$$

这个公式**主要用于 BJT 的 forward-active region**。深饱和区域不能用，边缘也可以用

导通边缘就是没有电流的情况

6.54 For the transistor shown in Fig. P6.54, assume $\alpha \simeq 1$ and $v_{BE} = 0.5$ V at the edge of conduction. What are the values of V_E and V_C for $V_B = 0$ V? For what value of V_B does the transistor cut off? Saturate? In each case, what values of V_E and V_C result?

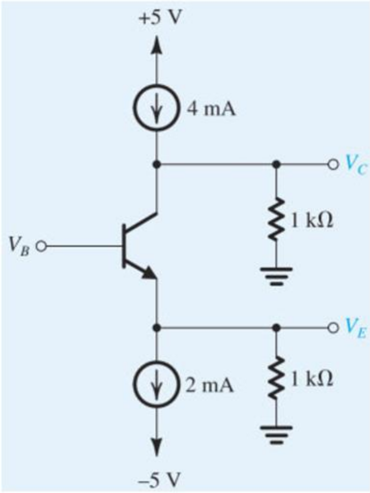


Figure P6.54

Problem 6.54

With $V_B = 0$ and active operation,

$$V_E = -0.7V.$$

The emitter node has a 2mA sink and a 1 $k\Omega$ resistor to ground. The transistor emitter current is

$$I_E = 2mA + \frac{V_E}{1k} = 2mA - 0.7mA = 1.3\,\text{mA}.$$

For large β, $I_C \approx 1.3$mA. The collector node receives 4mA, so the resistor current is $4 - 1.3 = 2.7$mA:

$$V_C = (2.7mA)(1k) = 2.7\,\text{V}.$$

At cutoff edge, the emitter current is zero. The emitter node is then set by the sink and resistor:

$$V_E = -2V.$$

Taking $V_{BE} \approx 0.5V$ at the edge gives

$$V_B \approx -1.5V.$$

At saturation edge, set $V_C = V_B$. Active expressions give

$$V_E = V_B - 0.7, \quad I_C \approx V_B + 1.3\,(mA), \quad V_C = 4 - I_C(1k) = 2.7 - V_B.$$

Thus

$$V_B = 2.7 - V_B \Rightarrow V_B = 1.35\,\text{V}.$$

知识点：电流源参与时，KCL 比“看图猜电流”可靠。先写节点电流，再判断工作区。

注意 KCL 的使用

D 6.57 Design a circuit using a *pnp* transistor for which $\alpha \simeq 1$ using two resistors connected appropriately to ±2 V so that $V_B = 0$ V, $I_E = 0.5$ mA, and $V_{BC} = 1$ V. What exact values of R_E and R_C would be needed? Now, consult the table of standard 5% resistor values provided in Appendix J to select suitable practical values. What values of resistors have you chosen? What are the values of I_E and V_{BC} that result?

Problem 6.57

For the pnp circuit, $V_B = 0$ and $I_E = 0.5\,\text{mA}$. With $V_{EB} \approx 0.7\text{V}$,

$$V_E = +0.7V.$$

Thus

$$R_E = \frac{2 - 0.7}{0.5mA} = 2.6\,\text{k}\Omega.$$

Given $V_{BC} = V_B - V_C = 1\text{V}$, the collector voltage is

$$V_C = -1V.$$

Thus

$$R_C = \frac{-1 - (-2)}{0.5mA} = 2.0\,\text{k}\Omega.$$

Using standard 5% values, choose

$$\boxed{R_E = 2.7\,k\Omega, \quad R_C = 2.0\,k\Omega.}$$

Then

$$I_E \approx \frac{2 - 0.7}{2.7k} = 0.481\,\text{mA}, \quad V_C = -2 + (0.481mA)(2k) = -1.04\,\text{V}.$$

知识点：标准阻值设计一定要计算“实际值”，不能只写理想电阻。

D *6.63 Design the circuit in Fig. P6.63 so that a current of 1 mA is established in the emitter and a voltage of −1 V appears at the collector. The transistor used has a nominal β of 100. However, the β value can be as low as 50 and as high as 150. Your design should ensure that the specified emitter current is obtained when β =100 and that at the extreme values of β the emitter current does not change by more than 10% of its nominal value. Also, design for as large a value for R_B as possible. Give the values of R_B, R_E, and R_C to the nearest kilohm. What is the expected range of collector current and collector voltage corresponding to the full range of β values?

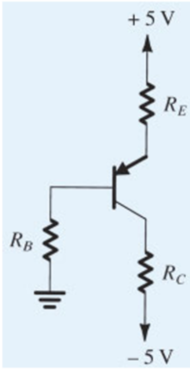


Figure P6.63

Problem 6.63

For the pnp design, target $I_E = 1\text{mA}$, $V_C = -1\text{V}$ at $\beta = 100$. Since

$$I_B = \frac{I_E}{\beta+1}, \quad V_B = I_B R_B, \quad V_E = V_B + 0.7,$$

the emitter-current equation is

$$I_E = \frac{5-(V_B+0.7)}{R_E} = \frac{4.3 - I_E R_B/(\beta+1)}{R_E}.$$

Equivalently,

$$I_E(\beta) = \frac{4.3}{R_E + R_B/(\beta+1)}.$$

At $\beta = 100$ and $I_E = 1\text{mA}$,

$$R_E + \frac{R_B}{101} = 4.3k\Omega.$$

To make R_B as large as possible while keeping I_E within ±10% over $50 \le \beta \le 150$, the lower-β end is limiting:

$$\frac{4.3k}{4.3k + R_B(1/51 - 1/101)} \ge 0.9.$$

Thus

$$R_B \le 49.2\,\text{k}\Omega.$$

Choose

$$R_B \approx 49\,\text{k}\Omega, \quad R_E = 4.3k - \frac{49k}{101} = 3.81\,\text{k}\Omega.$$

For the collector resistor,

$$I_C = \frac{100}{101}(1mA) = 0.990mA,$$
$$R_C = \frac{-1-(-5)}{0.990mA} = 4.04\,\text{k}\Omega.$$

Expected range:

β	$I_E(mA)$	$I_C(mA)$	$V_C(V)$
50	0.900	0.883	−1.43
100	1.000	0.990	−1.00
150	1.039	1.032	−0.83

$$\boxed{R_B \approx 49k\Omega,\ R_E \approx 3.81k\Omega,\ R_C \approx 4.04k\Omega}$$

知识点：要让偏置对 β 不敏感，需要让 emitter resistor 的电压降占主导；R_B 太大会让 base current 造成过大误差。

注意两端边界的使用

***6.66** For the circuit shown in Fig. P6.66, find the labeled node voltages for:

(a) $\beta = \infty$

(b) $\beta = 100$

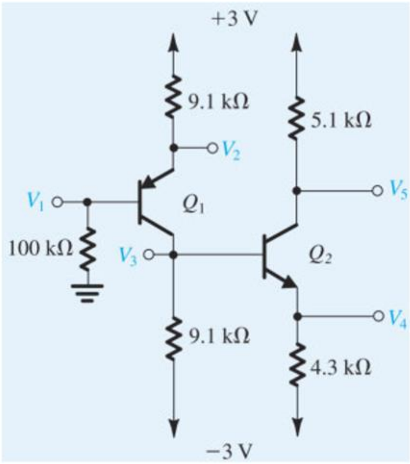


Figure P6.66

Problem 6.66

(a) $\beta = \infty$. For Q_1, $V_1 = 0$ and the pnp emitter is

$$V_2 = +0.7V, \quad I_{E1} = \frac{3 - 0.7}{9.1k} = 0.253mA.$$

Thus $I_{C1} \approx 0.253$mA. At node V_3,

$$\frac{V_3 + 3}{9.1k} = 0.253mA \Rightarrow V_3 = -0.7V.$$

For Q_2,

$$V_4 = V_3 - 0.7 = -1.4V, \quad I_{E2} = \frac{-1.4 + 3}{4.3k} = 0.372mA.$$

Then

$$V_5 = 3 - (0.372mA)(5.1k) = 1.10V.$$

(b) $\beta = 100$. For Q_1,

$$\frac{V_1}{100k} = \frac{[3 - (V_1 + 0.7)]/9.1k}{101}.$$

Solving,

$$V_1 = 0.226V, \quad V_2 = 0.926V, \quad I_{E1} = 0.228mA, \quad I_{C1} = 0.226mA.$$

At node V_3,

$$I_{C1} = \frac{V_3 + 3}{9.1k} + \frac{[(V_3 - 0.7) + 3]/4.3k}{101}.$$

Solving,

$$V_3 = -0.974V, \quad V_4 = -1.674V.$$

Then

$$I_{C2} = 0.305mA, \quad V_5 = 3 - (0.305mA)(5.1k) = 1.443V.$$

$$\boxed{\beta = \infty: \ V_1 = 0, V_2 = 0.7, V_3 = -0.7, V_4 = -1.4, V_5 = 1.10; \quad \beta = 100: \ V_1 = 0.226, V_2 = 0.926, V_3 = -0.974, V_4 = -1.674, V_5 = 1.443}$$

知识点： 多级 BJT 电路中，前一级 collector node 往往同时是后一级 base node；有限 β 时必须把后一级 base current 加入前一级节点 KCL。

多级 BJT，需要先算前面的 base，然后往回推，，要用 KCL 来联系多级，Vb 一直都是未知数

6.69 All the transistors in the circuits of Fig. P6.69(a), (b), (c) are specified to have a minimum β of 50. Find approximate values for the collector voltages and calculate forced β for each of the transistors. (**Hint**)

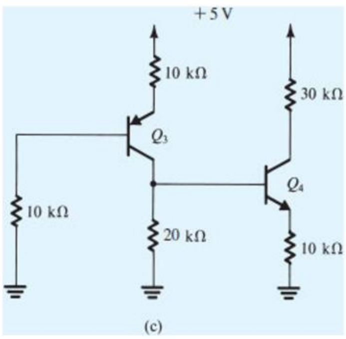


Figure P6.69(c)

(c) Let V_X be the node that is the collector of Q_3 and base of Q_4. Both transistors are saturated. For Q_3,

$$V_{E3} = V_{B3} + 0.7, \quad V_X = V_{E3} - 0.2 = V_{B3} + 0.5.$$

For Q_4,

$$V_{E4} = V_X - 0.7, \quad V_{C4} = V_{E4} + 0.2 = V_X - 0.5.$$

KCL at V_X gives the coupled solution

$$V_{B3} = 1.543V, \quad V_X = V_{C3} = 2.043V, \quad V_{E4} = 1.343V, \quad V_{C4} = 1.543V.$$

Currents are

$$I_{B3} = \frac{V_{B3}}{10k} = 0.154mA, \quad I_{C3} = 0.121mA, \quad \beta_{f3} = 0.786,$$
$$I_{C4} = \frac{5 - V_{C4}}{30k} = 0.115mA, \quad I_{B4} = 0.0191mA, \quad \beta_{f4} = 6.02.$$

All forced gains are well below 50, so the saturation assumption is consistent.

$$\boxed{(a)V_C = 0.804V,\ \beta_f = 2.27; \quad (b)V_C = 2.071V,\ \beta_f = 3.15; \quad (c)V_{C3} = 2.043V,\ V_{C4} = 1.543V,\ \beta_{f3} = 0.786,\ \beta_{f4} = 6.02}$$

知识点：饱和检查不能只说“电压看起来低”；要算 $\beta_f = I_C/I_B$，只要 $\beta_f < \beta_{min}$，就能保证最差晶体管也饱和。

注意 KCL 的使用

Chapter 7

7.10 A BJT amplifier circuit like the one in Fig. 7.6 is operated with V_{CC} = + 3 V and is biased at V_{CE} = +0.5 V. Find the voltage gain, the maximum allowed output negative swing without the transistor entering saturation, and the corresponding maximum input signal permitted.

∨ **Show Answer**

Problem 7.10

For the CE amplifier,

$$A_v = -g_m R_C = -\frac{I_C R_C}{V_T} = -\frac{V_{CC} - V_{CE}}{V_T}.$$

Thus

$$A_v = -\frac{3 - 0.5}{0.025} = -100\,\mathrm{V/V}.$$

The negative output swing is limited by saturation. Taking $V_{CEsat} \simeq 0.3\,\mathrm{V}$,

$$\hat{v}_{o,-} = 0.5 - 0.3 = 0.2\,\mathrm{V}, \qquad \hat{v}_{be} = 0.2/100 = 2\,\mathrm{mV}.$$

知识点： BJT CE 增益可直接写成 collector 电阻压降除以 V_T。

D 7.12 Consider the CE amplifier circuit of Fig. 7.6 when operated with a dc supply V_{CC} = +3 V. You want to find the point at which the transistor should be biased; that is, the value of V_{CE} so that the output sine-wave signal v_{ce} resulting from an input sine-wave signal v_{be} of 5-mV peak amplitude has the maximum possible magnitude. What is the peak amplitude of the output sine wave and the value of the gain obtained? Assume linear operation around the bias point. (**Hint**)

Problem 7.12

For v_{be} peak $= 5\,\mathrm{mV}$,

$$|\hat{v}_o| = |A_v|\hat{v}_{be} = \frac{3 - V_{CE}}{V_T}(5\,\mathrm{mV}) = 0.2(3 - V_{CE}).$$

The negative-swing limit is $V_{CE} - 0.3$. Maximum undistorted output occurs when the small-signal output equals this limit:

$$0.2(3 - V_{CE}) = V_{CE} - 0.3.$$

Thus $V_{CE} = 0.75\,\mathrm{V}$, $|\hat{v}_o| = 0.45\,\mathrm{V}$, and

$$A_v = -0.45/0.005 = -90\,\mathrm{V/V}.$$

知识点：最大输出不是盲目取最大增益；必须同时满足线性增益和摆幅边界。

注意这里的放大都是基于小信号的，增益的值也是小信号，也就是摇摆的部分

7.15 The transistor in the circuit of Fig. P7.15 is biased at a dc collector current of 0.2 mA. What is the voltage gain? (**Hint**)

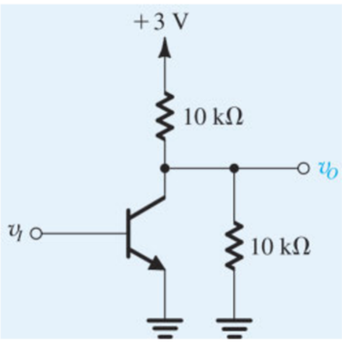


Figure P7.15

2. 小信号分析时，V_{CC} 怎么处理？

在 small-signal analysis 里，直流电源 $+3V$是 **AC ground**。

所以 collector 看到两个电阻都接到交流地：

$$10k\Omega \parallel 10k\Omega$$

因此小信号等效负载为：

$$R_{\text{load}} = 10k\Omega \parallel 10k\Omega$$

$$R_{\text{load}} = 5k\Omega$$

3. 求 g_m

BJT 小信号跨导：

$$g_m = \frac{I_C}{V_T}$$

取：

$$V_T \approx 25mV = 0.025V$$

代入：

$$g_m = \frac{0.2mA}{25mV}$$

$$g_m = \frac{0.0002}{0.025}$$

$$g_m = 0.008S$$

$$\boxed{g_m = 8mS}$$

4. 求电压增益

共射极放大器小信号增益：

$$A_v = -g_m R_{\text{load}}$$

代入：

$$A_v = -(8mS)(5k\Omega)$$

$$A_v = -(0.008)(5000)$$

$$\boxed{A_v = -40}$$

7.21 The purpose of this problem is to illustrate the application of graphical analysis to the circuit shown in Fig. P7.21. Sketch i_C–v_{CE} characteristic curves for the BJT for i_B = 10 µA, 20 µA, 30 µA, and 40 µA. Assume the lines to be horizontal (i.e., neglect the Early effect), and let β =100. For V_{CC} = 5 V and R_C = 1 kΩ, sketch the load line. What peak-to-peak collector voltage swing will result for i_B varying over the range 10 µA to 40 µA? If the BJT is biased at $V_{CE} = \frac{1}{2}V_{CC}$, find the value of I_C and I_B. If at this current V_{BE} = 0.7 V and if R_B = 100 kΩ, find the required value of V_{BB}.

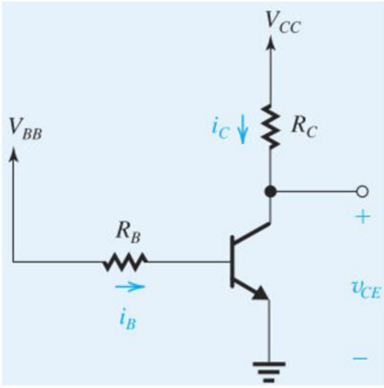


Figure P7.21

Problem 7.21

For horizontal BJT characteristics, $I_C = \beta I_B$. With $\beta = 100$, the four curves are at $1, 2, 3, 4$ mA. The load line is

$$i_C = \frac{5 - v_{CE}}{1k},$$

with intercepts $(v_{CE} = 5V, i_C = 0)$ and $(v_{CE} = 0, i_C = 5mA)$. As i_B varies from 10 to $40\,\mu A$, v_{CE} changes from

$$5 - (1mA)(1k) = 4V$$

to

$$5 - (4mA)(1k) = 1V.$$

Thus the collector swing is $3V_{p-p}$. If the desired midpoint is $I_C = 2.5mA$, then $I_B = 25\,\mu A$. With $V_{BE} = 0.7V$ and $R_B = 100k\Omega$,

$$V_{BB} = 0.7 + (25\mu A)(100k) = 3.2V.$$

知识点：图解分析本质上是 transistor characteristic 和 load line 的交点。

7.36 A transistor with β = 100 is biased to operate at a dc collector current of 0.4 mA. Find the values of g_m, r_π, and r_e. Repeat for a bias current of 40 µA.

∨ **Show Answer**

Problem 7.36

For $I_C = 0.4mA$,

$$g_m = 16mA/V, \quad r_\pi = \frac{100}{16mA/V} = 6.25k\Omega, \quad r_e = 62.5\Omega.$$

For $I_C = 40\mu A$,

$$g_m = 1.6mA/V, \quad r_\pi = 62.5k\Omega, \quad r_e = 625\Omega.$$

知识点： BJT 电流降低 10 倍，g_m 降低 10 倍，r_π 和 r_e 增大 10 倍。

Hybrid-π model 里的 r_π

$$\boxed{r_\pi = \frac{\beta}{g_m}}$$

T-model 里的 r_e

$$\boxed{r_e = \frac{1}{g_m}}$$

D 7.38 A designer wants to create a BJT amplifier with a g_m of 20 mA/V and a base input resistance of 4000 Ω or more. What collector-bias current should she choose? What is the minimum β she can tolerate for the transistor used?

∨ **Show Answer**

Problem 7.38

Required $g_m = 20mA/V$ gives

$$I_C = g_m V_T = 0.5mA.$$

The base input resistance is

$$r_\pi = \frac{\beta}{g_m} \geq 4k\Omega \Rightarrow \beta \geq (4k)(20mA/V) = 80.$$

知识点：想提高 g_m 必须提高偏置电流；想提高 r_π 则需要足够大的 β。

Base input resistance 是从 base 看进去对的电阻，不是真的在外面接一个电阻

7.39 A transistor operating with nominal g_m of 40 mA/V has a β that ranges from 60 to 150. Also, the bias circuit, being less than ideal, allows a ±10% variation in I_C. What are the extreme values found of the resistance looking into the base?

Problem 7.39

Since $r_\pi = \beta/g_m$ and $g_m = I_C/V_T$, the ±10% variation in I_C gives g_m from 36 to 44 mA/V. The minimum base resistance occurs with $\beta = 60$ and $g_m = 44mA/V$:

$$r_{\pi,min} = 1.36k\Omega.$$

The maximum occurs with $\beta = 150$ and $g_m = 36mA/V$:

$$r_{\pi,max} = 4.17k\Omega.$$

知识点： r_π 同时受 β 和偏置电流影响；极值要组合最不利条件。

D *7.41 We wish to design the amplifier circuit of Fig. 7.21 under the constraint that V_{CC} is fixed. Let the input signal $v_{be} = \hat{V}_{be} \sin \omega t$, where $\hat{V}_{be}$ is the maximum value for acceptable linearity. For the design that results in the largest signal at the collector, without the BJT leaving the active region, show that

$$R_C I_C = (V_{CC} - 0.3) \Big/ \left(1 + \frac{\hat{V}_{be}}{V_T}\right)$$

and find an expression for the voltage gain obtained. For $V_{CC} = 3$ V and $\hat{V}_{be} = 5$ mV, find the dc voltage at the collector, the amplitude of the output voltage signal, and the voltage gain.

注意观察结构

7.48 The transistor amplifier in Fig. P7.48 is biased with a current source I and has a very high β. Find the dc voltage at the collector, V_C. Also, find the value of r_e. Replace the transistor with the T model of Fig. 7.27(b) (note that the dc current source I should be replaced with an open circuit). Hence find the voltage gain v_c/v_i.

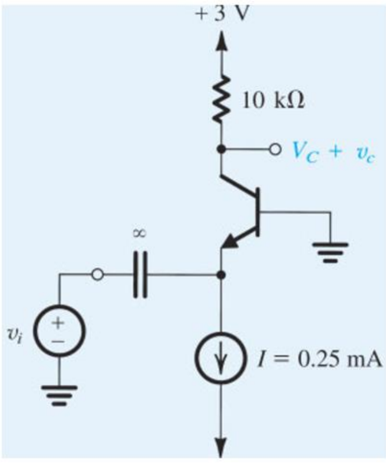


Figure P7.48

Problem 7.48

With very high β, $I_C \simeq I$. Hence

$$V_C = V_{CC} - IR_C$$

using the resistor shown in Fig. P7.48, and

$$r_e = \frac{V_T}{I_C} \simeq \frac{V_T}{I}.$$

Replacing the transistor by the T model and opening the dc current source, the gain is found by the voltage division between the signal resistance feeding the emitter and r_e, followed by the collector transresistance. For a simple common-base form,

$$\frac{v_c}{v_i} \simeq \frac{R_C}{r_e + R_{sig}}.$$

知识点： current-source bias fixes I_C and therefore fixes r_e。

注意这里的 high β

7.50 Figure P7.50 shows the circuit of an amplifier fed with a signal source v_{sig} with a source resistance R_{sig}. The bias circuitry is not shown. Replace the BJT with its hybrid-π equivalent circuit of Fig. 7.25(a). Find the input resistance $R_{in} \equiv v_\pi / i_b$, the voltage transmission from source to amplifier input, v_π / v_{sig}, and the voltage gain from base to collector, v_o / v_π. Use these to show that the overall voltage gain v_o / v_{sig} is given by

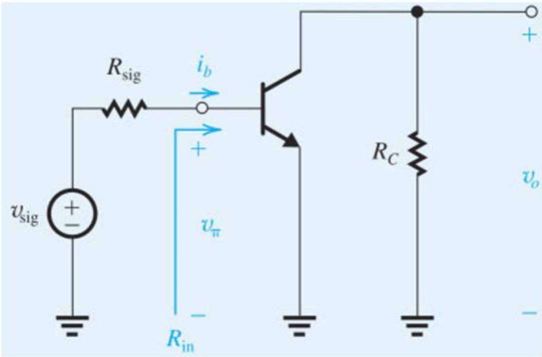


Figure P7.50

$$\frac{v_o}{v_{sig}} = -\frac{\beta R_C}{r_\pi + R_{sig}}$$

7.54 In the circuit shown in Fig. P7.54, the transistor has a β of 200. What is the dc voltage at the collector? Replacing the BJT with one of the hybrid-π models (neglecting r_o), draw the equivalent circuit of the amplifier. Find the input resistances R_{ib} and R_{in} and the overall voltage gain (v_o/v_{sig}). For an output signal of ±0.3 V, what values of v_{sig} and v_b are required?

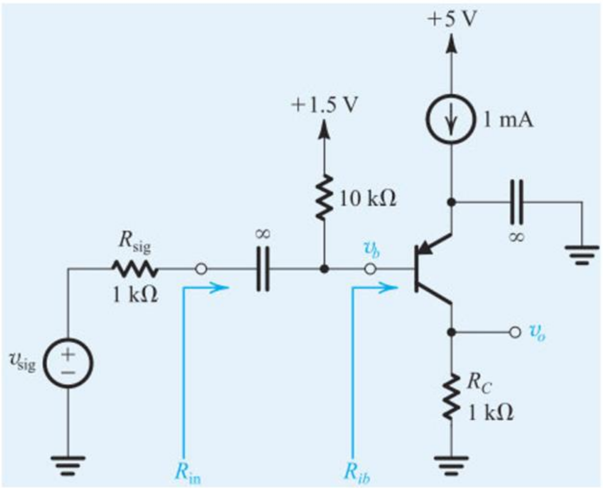


Figure P7.54

Problem 7.54

First solve the dc bias using $V_{BE} \simeq 0.7V$ and the stated $\beta = 200$. Then

$$g_m = I_C/V_T, \quad r_\pi = \beta/g_m.$$

For the ac equivalent, coupling capacitors are shorts and dc supplies are ac ground. The input resistance looking into the base is

$$R_{ib} = r_\pi + (\beta+1)R'_E$$

if an unbypassed emitter resistance is present. The total input resistance is $R_{in} = R_B \parallel R_{ib}$, and

$$\frac{v_o}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} A_v.$$

知识点： 带 emitter resistor 的 CE 放大器，输入电阻会被 $(\beta+1)$ 倍放大。

16.1 Consider MOS transistors fabricated in a 65-nm process for which $\mu_n C_{ox} = 540\ \mu\text{A/V}^2$, $\mu_p C_{ox} = 100\ \mu\text{A/V}^2$, $V_{tn} = -V_{tp} = 0.35$ V, and $V_{DD} = 1$V.

(a) Find R_{on} of an NMOS transistor with $W/L = 1.5$.

(b) Find R_{on} of a PMOS transistor with $W/L = 1.5$.

(c) If R_{on} of the PMOS device is to be equal to that of the NMOS device in (a), what must $(W/L)_p$ be?

Problem 16.1

The on-resistance of a MOS switch is evaluated from the triode-region slope at small v_{DS}:

$$R_{on} \simeq \frac{1}{k'(W/L)(V_{DD} - |V_t|)}.$$

For the NMOS device,

$$R_{on}, n = \frac{1}{(540\,\mu\text{A/V}^2)(1.5)(1 - 0.35)} = 1.90\,\text{k}\Omega.$$

For the equal-size PMOS,

$$R_{on}, p = \frac{1}{(100\,\mu\text{A/V}^2)(1.5)(0.65)} = 10.26\,\text{k}\Omega.$$

Equal resistance requires

$$\mu_p C_{ox}(W/L)_p = \mu_n C_{ox}(W/L)_n,$$

so

$$(W/L)_p = \frac{540}{100}(1.5) = 8.1.$$

$$\boxed{R_{on}, n = 1.90\,\text{k}\Omega, R_{on}, p = 10.26\,\text{k}\Omega, (W/L)_p = 8.1.}$$

知识点：PMOS 迁移率较低，所以同样尺寸时导通电阻更大；为了匹配 NMOS，PMOS 通常要做得更宽。

注意等效电阻的公式

和 MOSFET 线性区域的 rd 作区分

16.16 A particular logic inverter is specified to have $V_{IL} = 0.5$ V, $V_{IH} = 0.7$ V, $V_{OL} = 0.1$ V, and $V_{OH} = 1.2$ V. Find the high and low noise margins, NM_H and NM_L.

Problem 16.16

Noise margins are defined by

$$NM_L = V_{IL} - V_{OL}, \qquad NM_H = V_{OH} - V_{IH}.$$

Therefore

$$NM_L = 0.5 - 0.1 = 0.4\,\text{V}, \qquad NM_H = 1.2 - 0.7 = 0.5\,\text{V}.$$

$$\boxed{NM_L = 0.4\,\text{V}, \quad NM_H = 0.5\,\text{V}.}$$

知识点：低噪声容限看 V_{IL} 和 V_{OL} 的间距；高噪声容限看 V_{OH} 和 V_{IH} 的间距。

16.17 The voltage-transfer characteristic of a particular logic inverter is modeled by three straight-line segments in the manner shown in Fig. 16.13. If $V_{IL} = 1.1$ V, $V_{IH} = 1.2$ V, $V_{OL} = 0.3$ V, and $V_{OH} = 1.5$ V, find:

(a) the noise margins

(b) the value of V_M

(c) the voltage gain in the transition region

Problem 16.17

The noise margins are

$$NM_L = 1.1 - 0.3 = 0.8\,\text{V}, \qquad NM_H = 1.5 - 1.2 = 0.3\,\text{V}.$$

The transition segment passes through $(V_{IL}, V_{OH}) = (1.1, 1.5)$ and $(V_{IH}, V_{OL}) = (1.2, 0.3)$, so its slope is

$$A_v = \frac{0.3 - 1.5}{1.2 - 1.1} = -12\,\text{V/V}.$$

Using $v_O = 1.5 - 12(v_I - 1.1)$ and setting $v_O = v_I$,

$$V_M = 1.5 - 12(V_M - 1.1) \Rightarrow 13V_M = 14.7,$$

$$\boxed{V_M = 1.13\,\text{V}, \quad A_v = -12\,\text{V/V}.}$$

知识点：三段线性 VTC 中，V_M 是转移线和 $v_O = v_I$ 的交点。

注意是反向的输出，要记住如何绘制线段

放大的倍率就是斜率，Vm 是转折点，y=x 与线段的交点

16.28 Consider a CMOS inverter fabricated in a 65-nm CMOS process for which $V_{DD} = 1$ V, $V_{tn} = -V_{tp} = 0.35$ V, and $\mu_n C_{ox} = 5.4\mu_p C_{ox} = 540\ \mu A/V^2$. In addition, Q_N and Q_P have $L = 65$ nmand $(W/L)_n = 1.5$.

(a) Find W_p that results in $V_M = V_{DD}/2$. What is the silicon area utilized by the inverter in this case?

(b) For the matched case in (a), find the values of V_{OH}, V_{OL}, V_{IH}, V_{IL}, NM_L, and NM_H.

(c) For the matched case in (a), find the output resistance of the inverter in each of its two states.

Problem 16.28

For $V_M = V_{DD}/2$ with equal thresholds, match the transconductance parameters:

$$\mu_n C_{ox}(W/L)_n = \mu_p C_{ox}(W/L)_p.$$

Since $\mu_n C_{ox} = 5.4\mu_p C_{ox}$,

$$(W/L)_p = 5.4(1.5) = 8.1.$$

With $L = 65$ nm,

$$W_n = 1.5(65) = 97.5\,\text{nm}, \quad W_p = 8.1(65) = 526.5\,\text{nm}.$$

The silicon gate area is approximately

$$A = L(W_n + W_p) = 65(97.5 + 526.5)\,\text{nm}^2 = 0.0406\,\mu\text{m}^2.$$

For the matched case,

$$V_{OH} = 1\,\text{V}, \quad V_{OL} = 0,$$

$$V_{IH} = \frac{5(1) - 2(0.35)}{8} = 0.5375\,\text{V}, \quad V_{IL} = \frac{3(1) + 2(0.35)}{8} = 0.4625\,\text{V}.$$

Thus

$$NM_L = NM_H = 0.4625\,\text{V}.$$

The output resistance in either state is the corresponding on-resistance:

$$R_{out,L} = \frac{1}{(540\mu)(1.5)(0.65)} = 1.90\,\text{k}\Omega,$$

注意 a 的设问背后的含义是二者的 k 一致

面积是二者加一起

$$R_{out,H} = \frac{1}{(100\mu)(8.1)(0.65)} = 1.90\,\text{k}\Omega.$$

知识点： Matched CMOS 让 VTC 对称，因此 $V_M = V_{DD}/2$ 且两个 noise margin 相等。

$$V_{IL} = \frac{3V_{DD} + 2V_t}{8}, V_{IH} = \frac{5V_{DD} - 2V_t}{8}$$

公式注意使用条件，要是 Vm = Vdd / 2, VIL 和 VIH 也是关于这个对称的

16.31 A CMOS inverter for which $k_n = 5k_p = 250\ \mu A/V^2$ and $V_t = 0.4$ V is connected as shown in Fig. P16.31 to a sinusoidal signal source having a Thévenin equivalent voltage of 0.1-V peak amplitude and resistance of 100 kΩ. What signal voltage appears at node A with $v_I = +1.5$ V? With $v_I = -1.5$ V?

∨ **Show Answer**

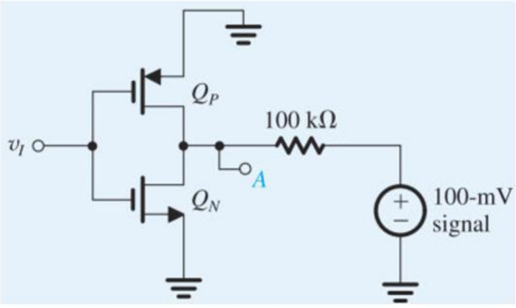


Figure P16.31

Problem 16.31

The signal source sees the on-resistance of whichever transistor is turned on by v_I. For $v_I = +1.5$ V, the NMOS is on:

$$R_{on,n} = \frac{1}{k_n(V_{GS} - V_t)} = \frac{1}{(250\mu)(1.5 - 0.4)} = 3.64\,\text{k}\Omega.$$

The signal at node A is the divider result:

$$\hat{v}_A = 0.1\frac{3.64k}{100k + 3.64k} = 3.51\,\text{mV}.$$

For $v_I = -1.5$ V, the PMOS is on. Since $k_n = 5k_p$, $k_p = 50\,\mu\text{A}/\text{V}^2$, and

$$R_{on,p} = \frac{1}{(50\mu)(1.1)} = 18.18\,\text{k}\Omega.$$

Then

$$\hat{v}_A = 0.1\frac{18.18k}{100k + 18.18k} = 15.4\,\text{mV}.$$

知识点：这题把 CMOS inverter 当作受控制的开关电阻；哪个管导通，就用哪个管的 R_{on} 做分压。

注意这一题的意思，需要读懂题意，相当于一个单刀双置的开关

16.33 Consider the CMOS inverter of Fig. 16.22 with Q_N and Q_P matched and with the input v_I rising slowly from 0 to V_{DD}. At what value of v_I does the current flowing through Q_N and Q_P reach its peak? Give an expression for the peak current, neglecting λ_n and λ_p. For k_n' = 540 μA/V^2, $(W/L)_n$ = 1.5, V_{DD} = 1.0 V, and V_{tn} = 0.35V, find the value of the peak current.

3. 为什么峰值电流出现在 $v_I = V_{DD}/2$？

当 v_I较低时：

$$v_{GS,n} = v_I$$

NMOS 的 overdrive 是：

$$v_I - V_t$$

比较小，所以 NMOS 较弱。

当 v_I较高时：

$$v_{SG,p} = V_{DD} - v_I$$

PMOS 的 overdrive 是：

$$V_{DD} - v_I - |V_t|$$

比较小，所以 PMOS 较弱。

中间点：

$$v_I = \frac{V_{DD}}{2}$$

时，NMOS 和 PMOS 的 overdrive 相等：

$$v_I - V_t = V_{DD} - v_I - V_t$$

所以两管同时比较强，而且强度匹配，因此直通电流最大。

解这个等式：

$$v_I - V_t = V_{DD} - v_I - V_t$$

$$2v_I = V_{DD}$$

$$\boxed{v_I = \frac{V_{DD}}{2}}$$

Problem 16.33

For a matched CMOS inverter, the largest short-circuit current occurs at the symmetric switching point

$$\boxed{v_I = \frac{V_{DD}}{2}}.$$

There both devices are in saturation and

$$I_{peak} = \frac{1}{2}k_n\left(\frac{V_{DD}}{2} - V_t\right)^2.$$

Here

$$k_n = k_n'(W/L)_n = (540\,\mu)(1.5) = 810\,\mu\mathrm{A/V}^2.$$

Thus

$$I_{peak} = \frac{1}{2}(810\,\mu)(0.5 - 0.35)^2 = 9.11\,\mu\mathrm{A}.$$

知识点： Matched CMOS 的直流贯通电流在 $v_I = V_{DD}/2$ 附近最大。

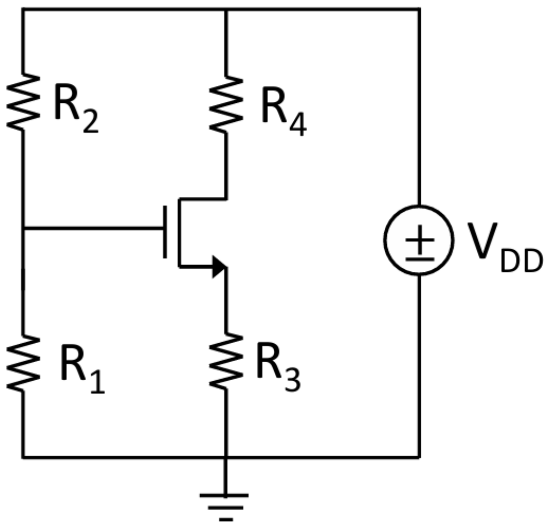
CMOS 中 是 VDD-PMOS-NMOS-GND

所以是串联的，算了一个的电流就是全路的电流

3. The NMOS FET has K_N'=100μA/V^2 and V_{tn}=0.75V and the circuit includes R_1 = 1000 kΩ, R_2 = 2200 kΩ, R_3 = 240 kΩ, R_4 = 120 kΩ, and V_{DD} = 10V. Assume that K_N'=100μA/V^2, V_{tn}=0.75V and W/L=6/1. (**20 points**).

   a. Find the Q-point (V_{GS}, V_{DS}, I_D) for the transistor if W/L = 6.

   b. Repeat (a) for W/L=12/1.



1. 先求 Gate 电压

Gate 电流近似为 0，所以 R_2, R_1是普通分压器：

$$V_G = V_{DD}\frac{R_1}{R_1 + R_2}$$

代入：

$$V_G = 10\frac{1000}{1000 + 2200} = 3.125V$$

$$\boxed{V_G = 3.125V}$$

2. 写出各节点电压

电流 I_D流过 R_3和 R_4。

Source 电压：

$$V_S = I_D R_3$$

Drain 电压：

$$V_D = V_{DD} - I_D R_4$$

所以：

$$V_{GS} = V_G - V_S = 3.125 - I_D R_3$$

$$V_{DS} = V_D - V_S = 10 - I_D (R_4 + R_3)$$